

Esercizi

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1. *Risolvere le seguenti equazioni goniometriche:*

(a) $\sin(3x + \pi) = 0$

(d) $\cos\left(\frac{\pi}{12} - x\right) = 1$

(b) $2 \sin\left(3x + \frac{\pi}{3}\right) - \sqrt{3} = 0$

(e) $2 \sin 2x - \sqrt{3} = 0$

(c) $\cos\left(2x + \frac{\pi}{6}\right) = -\frac{1}{2}$

(f) $\tan\left(3x + \frac{\pi}{6}\right) = -1$

2. *Risolvere le seguenti equazioni goniometriche:*

(a) $\sin^2 x + 3 \sin x + 3 = 0$

(e) $\left(\sin x - \sqrt{2}\right)^2 = \frac{1}{2}$

(b) $2 \cos^2 x - 3 \cos x + 1 = 0$

(f) $\left(\cos x - \frac{1}{2}\right)^2 = \frac{1}{4}$

(c) $2 \tan^2 x - \tan x - 1 = 0$

(g) $\sin x(2 \sin x + \sqrt{2}) = 2 \sin x + \sqrt{2}$

(d) $2 \tan^2 x = \tan x$

(h) $(2 \cos x - 1) \cos x = 2 \cos x - 1$

3. *Risolvere le seguenti equazioni goniometriche:*

(a) $\sin^x - \cos x + 1 = 0$

(f) $\sin 2x = \sqrt{2} \cos x$

(b) $2 \sin^2 x + \sin(\pi + x) - 1 = 0$

(g) $\cos 2x - \sin x = -2$

(c) $\sin^2 x - \cos x - 1 = 0$

(h) $4 \sin x \cos x = \sqrt{2}$

(d) $2 \cos^2 x - \sin\left(\frac{\pi}{2} + x\right) - 1 = 0$

(e) $6 \cos^2 x - \cos\left(x + \frac{\pi}{2}\right) - 5 = 0$

(i) $\sin 2x = 2 \cos x$